

Use Case Document

Solari — AI-Powered Solar Industry Chatbot

Track 2 — Project Group AI_02

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This one walks through who we actually built Solari for, and what those people are doing with it day to day — written from our own understanding of the users, not a boilerplate list.

1. Purpose

This document is about the people Solari is actually for, and the real things they'd want to do with it. It pairs with the Requirements Document — where that one says what the system must do, this one shows why, by walking through each requirement from the point of view of someone actually sitting at the keyboard asking a question.

2. Target Users

2.1 Prospective Residential Customer

Picture a homeowner looking into solar for the first time. They probably don't know the difference between a kW and a kWh, and they're not going to pretend they do. What they want is a rough sense of size and cost, and honestly, most of them just want to know if their bill is actually going to go down under Pakistan's Net-Billing rules.

2.2 Commercial / Industrial Customer

This is more of a business owner or a facility manager sizing something up for a bigger, non-residential load. They tend to ask sharper questions — on-grid vs. hybrid vs. off-grid, how much backup they'll get during load-shedding, and the DISCO interconnection rules that a homeowner would never think to ask about, like the 3-phase requirement or the 80% transformer rule.

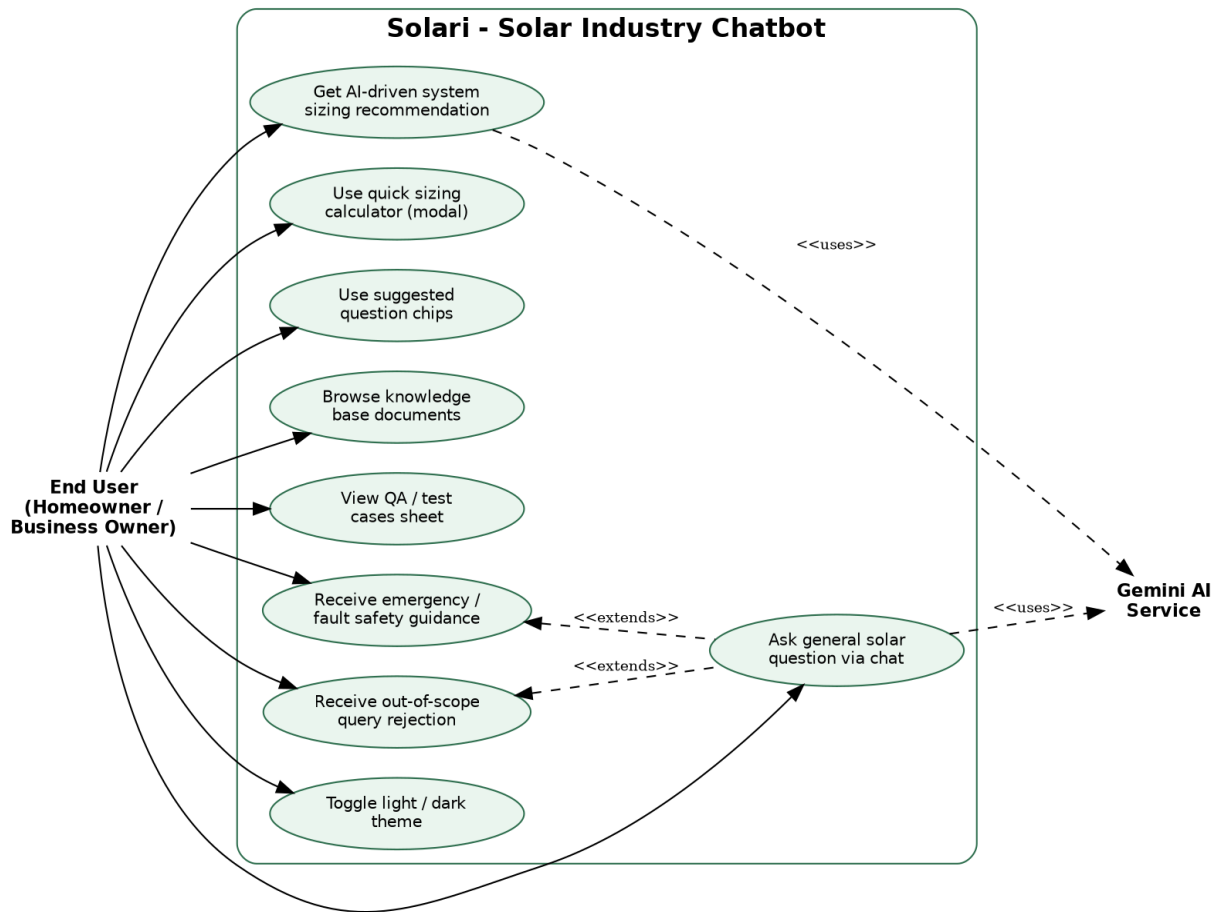
2.3 Existing Solar System Owner

Someone who already has a system running and just needs day-to-day operational help — when to clean the panels, what a particular fault code means, or what to actually do if something starts sparking or smelling like it's burning.

2.4 Project Evaluator

And then there's us, or whoever's reviewing this — a team member, instructor, or evaluator going through the knowledge base and the recorded test cases to check the assistant is actually behaving the way it's supposed to and staying inside its lane.

3. Use Case Diagram



Almost everything on this diagram traces back to the End User — they're the ones driving the interaction, every time. The two safety-related use cases (emergency guidance and out-of-scope rejection) aren't things a user asks for directly; they just extend the regular chat flow whenever the conversation happens to need them. And both of the AI-driven paths lean on the external Gemini service to actually generate a response.

4. Use Case Summary

ID	Use Case	Primary Actor
UC-01	Ask a general solar question via chat	End User
UC-02	Get an AI-driven system sizing recommendation	End User
UC-03	Use the quick sizing calculator (modal)	End User
UC-04	Use a suggested question chip	End User
UC-05	Browse knowledge base documents	End User / Evaluator
UC-06	View the QA / test cases sheet	Evaluator
UC-07	Receive emergency / fault safety guidance	End User
UC-08	Receive an out-of-scope query rejection	End User
UC-09	Toggle light / dark theme	End User

5. Detailed Use Cases

UC-01 — Ask a General Solar Question via Chat

Use Case ID	UC-01
Name	Ask a general solar question via chat
Actor(s)	End User
Description	The user asks Solari an open-ended question about panels, inverters, batteries, installation, or Pakistan's net-billing rules, and receives an answer grounded in the knowledge base.
Preconditions	• The chat page is loaded and the backend is reachable. • A valid Gemini API key is configured on the backend.
Main Flow	1. User types a question in the composer and presses Send or Enter. 2. Frontend disables the composer and shows a typing indicator. 3. Backend sends the message, knowledge base, and system prompt to the Gemini API. 4. AI generates a grounded answer in Markdown. 5. Frontend renders the sanitized Markdown reply as a chat bubble. 6. Composer re-enables for the next message.
Alternate Flow	• If the request fails or times out, the frontend shows a network-error banner and keeps the user's message in the thread. • If the question implies a sizing request, the flow proceeds instead as UC-02.
Postconditions	• The chat thread contains the user's question and the assistant's grounded reply.

UC-02 — Get an AI-Driven System Sizing Recommendation

Use Case ID	UC-02
Name	Get an AI-driven system sizing recommendation
Actor(s)	End User, Gemini AI Service
Description	The user asks the chat assistant to size a solar system for them (e.g. “I use 500 units a month, what size do I need?”) and the assistant invokes the deterministic sizing tool rather than estimating the numbers itself.
Preconditions	• The user has provided or is asked for their monthly electricity usage. • The Gemini API and tool-calling configuration are available.
Main Flow	1. User sends a message describing their usage and, optionally, backup needs and system type. 2. AI recognizes a sizing intent and calls the <code>calculate_solar_recommendation</code> tool with the extracted arguments. 3. Backend executes the deterministic sizing formulas and returns system kW, panel count, roof area, and battery capacity.

	- AI composes a natural-language explanation of the result and appends the mandatory disclaimer. - Frontend displays the explanation in the chat thread.
Alternate Flow	- If the monthly usage was not provided, the assistant asks a clarifying follow-up instead of calling the tool. - If the provided values are invalid, the backend returns an error message explaining what to correct.
Postconditions	The user has a system size, panel count, roof area, and battery estimate, with the mandatory disclaimer attached.

UC-03 — Use the Quick Sizing Calculator (Modal)

Use Case ID	UC-03
Name	Use the quick sizing calculator (modal)
Actor(s)	End User
Description	The user opens the standalone calculator modal to get an instant sizing estimate without going through the conversational flow.
Preconditions	The chat page is loaded.
Main Flow	- User clicks “Calculate Now” in the side panel to open the calculator modal. - User enters average monthly usage (kWh) and, optionally, required backup hours. - User submits the form. - Frontend calls POST /api/recommend directly (no AI call). - Backend returns the deterministic recommendation. - Modal displays the recommended system size, panel count, roof area, and battery capacity.
Alternate Flow	- If monthly usage is left blank or invalid, the browser-native form validation blocks submission. - User can close the modal at any point via the close button.
Postconditions	The user sees an instant sizing result without needing to converse with the AI.

UC-04 — Use a Suggested Question Chip

Use Case ID	UC-04
Name	Use a suggested question chip
Actor(s)	End User
Description	A first-time user taps a pre-written prompt (“Size my system”, “Compare panels”, “Installation process”, “Talk to a person”) instead of typing a question from scratch.
Preconditions	The chat page is loaded and no message is currently being sent.

Main Flow	1. User taps a suggestion chip. 2. The chip's text is sent as a chat message automatically, following the same flow as UC-01.
Alternate Flow	• If a message is already in flight, chip taps are ignored until the current request completes.
Postconditions	• The chosen prompt appears in the thread along with the assistant's reply.

UC-05 — Browse Knowledge Base Documents

Use Case ID	UC-05
Name	Browse knowledge base documents
Actor(s)	End User / Evaluator
Description	The user reviews the source PDF documents that ground the assistant's answers, to verify or explore the underlying research.
Preconditions	• The knowledge base PDFs are present on the backend.
Main Flow	1. User clicks “Knowledge base” in the side navigation. 2. Frontend switches to the knowledge base view listing each source document. 3. User clicks a document card to open the PDF in a new browser tab.
Alternate Flow	• None.
Postconditions	• The user has viewed or downloaded a knowledge base source document.

UC-06 — View the QA / Test Cases Sheet

Use Case ID	UC-06
Name	View the QA / test cases sheet
Actor(s)	Project Evaluator
Description	An evaluator opens the recorded test case spreadsheet to review how the assistant's knowledge-base grounding was validated.
Preconditions	• The test case spreadsheet is present on the backend.
Main Flow	1. User clicks “Test cases” in the side navigation. 2. Frontend switches to the test cases view. 3. User clicks the spreadsheet card to open/download the file.
Alternate Flow	• None.
Postconditions	• The evaluator has access to the full set of recorded test cases.

UC-07 — Receive Emergency / Fault Safety Guidance

Use Case ID	UC-07
Name	Receive emergency / fault safety guidance
Actor(s)	End User
Description	A user reports a dangerous symptom (sparking, burning smell, smoke, or a persistent fault code) and the assistant must respond with the mandatory safety protocol instead of general advice.
Preconditions	The user's message contains an emergency indicator recognized by the system prompt.
Main Flow	- User describes the symptom in chat (e.g. "my inverter is smoking"). - AI matches the emergency protocol defined in the system prompt. - AI responds with the fixed shutdown sequence: turn off the DC isolator, turn off the main AC breaker, do not touch the wires, contact a certified installer.
Alternate Flow	This use case extends UC-01 — it is triggered mid-conversation whenever an emergency indicator appears.
Postconditions	The user has explicit, immediate safety instructions rather than a general explanation.

UC-08 — Receive an Out-of-Scope Query Rejection

Use Case ID	UC-08
Name	Receive an out-of-scope query rejection
Actor(s)	End User
Description	A user asks something unrelated to solar energy (e.g. cooking, sports, general coding) and the assistant declines while restating its domain.
Preconditions	None.
Main Flow	- User sends an off-domain message. - AI recognizes the request falls outside solar PV, inverter backup, energy management, or net-billing regulation. - AI replies with the fixed scope-decline statement.
Alternate Flow	This use case extends UC-01 — it is triggered whenever a message falls outside the assistant's defined domain.
Postconditions	The user understands the assistant's scope and can redirect their question.

UC-09 — Toggle Light / Dark Theme

Use Case ID	UC-09
Name	Toggle light / dark theme
Actor(s)	End User
Description	The user switches the interface's visual theme for comfort or preference.

Preconditions	The chat page is loaded.
Main Flow	- User clicks the theme toggle button in the header. - Frontend switches the active theme immediately.
Alternate Flow	None.
Postconditions	The interface reflects the newly selected theme for the remainder of the session.